

INDU 211: Introduction to Production & Manufacturing Systems

Problem Solution Guide 04: Queuing Theory ($M/M/1$) System Evaluation

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[!NOTE]

- **YouTube Video Tutorial:** [Watch Video 4: INDU 211 - Queuing Theory Problem](#)
- **Video ID:** XT1EgQRcqmU · **Duration:** 17:23
- **Target Exam Scope:** 🎯 **FINAL EXAM (Weeks 7–12, Chapters 14, 15, 8, 17)**
- **Curriculum Context:** Chapter 15 — Probabilistic Operations Research Models

1. Operating Parameters & Model Classification

A downtown takeout fast-food restaurant operates a single customer service window open daily from 11:00 AM to 11:00 PM:

- **Daily Operating Horizon:** $T = 12$ hours/day (720 minutes).
- **Customer Arrival Distribution:** Poisson process with an average inter-arrival time of 10 minutes:

$$\text{Mean Arrival Rate } \lambda = \frac{1 \text{ customer}}{10 \text{ minutes}} = \mathbf{6 \text{ customers/hour}}$$

- **Customer Service Distribution:** Exponential service process with an average service time of 4 minutes (order entry, preparation, payment):

$$\text{Mean Service Rate } \mu = \frac{60 \text{ minutes/hour}}{4 \text{ minutes/customer}} = \mathbf{15 \text{ customers/hour}}$$

- **Queuing Model Classification:** Single server ($s = 1$), Poisson arrivals (M), Exponential service times (M), infinite queue capacity, First-Come-First-Served (FCFS) discipline: $M/M/1$ **Queuing System**.

2. Step-by-Step Worked Solutions

Part 5.1: Estimated Server Busy Time in the 12-Hour Day

- **Step 1: Calculate Traffic Intensity (Server Utilization Factor ρ):**

$$\rho = \frac{\lambda}{\mu} = \frac{6}{15} = \mathbf{0.40} \quad (40)$$

(Because $\rho = 0.40 < 1.0$, the queuing system reaches steady-state stability).

- **Step 2: Calculate Daily Window Busy Time:**

$$\text{Daily Busy Time} = \rho \times T = 0.40 \times 12 \text{ hours} = \mathbf{4.80 \text{ hours/day}}$$

$$\text{In Minutes} = 4.80 \times 60 = \mathbf{288} \text{ minutes/day}$$

Part 5.2: Expected Customer Waiting Time in Queue (W_q)

- **Formula:**

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)}$$

- **Calculation:**

$$W_q = \frac{6}{15(15 - 6)} = \frac{6}{15 \times 9} = \frac{6}{135} = \frac{2}{45} \text{ hours} \approx 0.0444 \text{ hours}$$

- **Convert to Minutes:**

$$W_q = \frac{2}{45} \times 60 = \frac{120}{45} = \mathbf{2.67} \text{ minutes}$$

- **Interpretation:** An arriving customer waits in line an average of **2 minutes and 40 seconds** before reaching the service window.
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Part 5.3: Expected Total Time in the System (W)

- **Formula:**

$$W = W_q + \frac{1}{\mu} = \frac{1}{\mu - \lambda}$$

- **Calculation:**

$$W = \frac{1}{15 - 6} = \frac{1}{9} \text{ hours} \approx 0.1111 \text{ hours}$$

- **Convert to Minutes:**

$$W = \frac{1}{9} \times 60 = \mathbf{6.67} \text{ minutes}$$

- **Interpretation:** A customer spends an average of **6 minutes and 40 seconds** total at the restaurant (waiting in line plus ordering, paying, and receiving food).
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Part 5.4: Congestion Analysis — Daily Hours with ≥ 2 Customers in the System

- **Step 1: State Probability for an $M/M/1$ Queue:** The steady-state probability of having exactly n customers in the system is:

$$P_n = (1 - \rho)\rho^n$$

- **Step 2: Probability of 2 or More Customers ($P(n \geq 2)$):**

$$P(n \geq 2) = 1 - P_0 - P_1 = 1 - (1 - \rho)\rho^0 - (1 - \rho)\rho^1 = 1 - (1 - \rho) - \rho(1 - \rho) = \rho - \rho(1 - \rho) = \rho^2$$

- **Substitute Utilization Factor** $\rho = 0.40$:

$$P(n \geq 2) = (0.40)^2 = \mathbf{0.16} \quad (16)$$

- **Step 3: Calculate Daily Congestion Operating Hours:**

$$\text{Daily Congestion Time} = P(n \geq 2) \times T = 0.16 \times 12 \text{ hours} = \mathbf{1.92} \text{ hours/day}$$

$$\text{In Minutes} = 1.92 \times 60 = \mathbf{115.2} \text{ minutes/day}$$

- **Conclusion:** During a typical 12-hour business day, the restaurant experiences a queue with two or more customers for approximately **1 hour and 55 minutes**.